Kousumi Mondal

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Professional Summary

Final year B.Tech Computer Science & Engineering student with hands-on experience developing end-to-end ML pipelines and production-grade RAG systems. Skilled in building scalable AI solutions across NLP, computer vision, and LLM integration, with a strong focus on generative AI, semantic retrieval, and backend system design.

Education

Vellore Institute of Technology Bhopal, India 2023–2027
B.Tech Computer Science & Engineering – CGPA: 8.07

Ideal Public School Howrah, India 2021–2023
ISC Class XII (Science) – 74.98%

Ideal Public School Howrah, India 2013–2021
ICSE Class X – 87.4%

Technical Skills

Languages & Core: Python, Java, SQL, Git, Linux
Machine Learning & NLP: scikit-learn, LangChain, Sentence Transformers, Vector Search, RAG Pipelines, LLM Integration
Data Processing & Visualization: Pandas, NumPy, Matplotlib, Seaborn, Hadoop (Fundamentals)
Backend & Databases: FastAPI, REST APIs, MongoDB, MySQL, PostgreSQL

Projects

DRUNet Image Denoiser GitHub | Live

- Developed a DRUNet-based image denoising model on the DIV2K dataset, achieving high-quality image restoration through deep residual learning techniques.
- Built an end-to-end training pipeline incorporating realistic noise injection, data augmentation, and optimization strategies for robust model generalization.
- Evaluated performance using PSNR and SSIM metrics and optimized inference workflows for efficient processing of high-resolution images.

LexTrace – Legal Document Intelligence System GitHub | Live

- Engineered a full-stack RAG pipeline for legal document ingestion and querying, leveraging PyMuPDF and Tesseract OCR to process both scanned and digital legal documents.
- Implemented semantic retrieval using Sentence Transformers and MongoDB vector search, enabling context-aware retrieval across large legal corpora with reduced query latency.
- Integrated a Groq-hosted LLM through LangChain for citation-aware legal question answering and developed a responsive TypeScript frontend for seamless user interaction.

MedInsight – Medical AI Intelligence Platform GitHub | Live

- Built a medical AI platform combining LangChain, Groq-hosted LLMs, and ChromaDB to enable retrieval-augmented clinical question answering over biomedical literature.
- Designed semantic retrieval pipelines for research paper processing, vector indexing, and context-aware response generation using biomedical data sources.
- Developed a FastAPI backend and JavaScript frontend while leveraging Biopython for structured parsing and extraction of biological research information.

Achievements

Meta X PyTorch OpenEnv Hackathon – Finalist 2026
Selected among 800 finalist teams from 31,000+ registered participants nationwide.